

Black holes

Geometry, thermodynamics, and quantum information

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A black hole is first a causal region, then a thermodynamic system, and finally a stringent test of quantum mechanics. These descriptions are connected, but their assumptions should not be collapsed into a single picture.

This note develops concrete calculations in Schwarzschild and anti-de Sitter geometries, explains the thermal origin of Hawking radiation, and follows the information problem toward quantum extremal surfaces. The microscopic string-theory example is explicitly distinguished from generic astrophysical black holes.

The prerequisites are classical mechanics, special relativity, differential geometry, and introductory quantum field theory. Units are $c = \hbar = k_B = 1$ unless constants are explicitly restored. Standard mass, radius, entropy, and temperature notation is retained throughout.

These are pedagogical study notes, with worked calculations and reading references. They summarize established results and indicate the assumptions behind semiclassical or conjectural statements.

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1 A horizon is a causal statement

In an asymptotically flat spacetime, the black-hole region consists of events that cannot send future-directed causal signals to future null infinity. Its boundary is the event horizon. This definition depends on the full future development of the spacetime: it cannot generally be identified from a local curvature measurement.

A trapped surface has both future-directed null expansions negative. Apparent and trapping horizons are useful quasi-local notions, but they depend on choices such as spacetime slicing and need not coincide with the event horizon in a dynamical geometry. In stationary solutions these distinctions simplify, which is why equilibrium black holes are effective starting points.

A Killing horizon is a null hypersurface generated by a Killing vector that becomes null there. The familiar thermodynamic formulas apply most directly to stationary Killing horizons with an appropriate normalization of the generator. A generic dynamical horizon does not have a unique equilibrium temperature obtained by copying the stationary formula.

The equivalence principle is not violated when an observer crosses a sufficiently large, smooth horizon. Curvature can be small there. The inability to return is a statement about the global orientation of future light cones, not a material wall at the horizon.

Check. A large redshift by itself does not establish the existence of an event horizon. One must ask whether outgoing causal curves can ultimately reach the relevant asymptotic boundary.

Reading connection: For the causal and geometric foundations, consult Tong's general-relativity lectures [1] and Reall's black-hole lectures [2].

2 Schwarzschild geometry and curvature

Outside a spherical, nonrotating vacuum source,

$$ds^2 = - \left(1 - \frac{2GM}{r} \right) dt^2 + \frac{dr^2}{1 - 2GM/r} + r^2 d\Omega_2^2. \quad (2.1)$$

The coordinate r is an area radius: a symmetry sphere has area $4\pi r^2$. This geometric meaning makes it particularly useful for locating the horizon at $r = 2GM$.

The apparent divergence in the radial metric coefficient is not a curvature singularity. A scalar curvature invariant is

$$R_{abcd}R^{abcd} = \frac{48G^2M^2}{r^6}. \quad (2.2)$$

It is finite at the horizon and diverges at $r = 0$. Vacuum Einstein equations give $R_{ab} = 0$, but they do not require the full Riemann tensor to vanish.

A static clock at radius r measures proper time $d\tau = \sqrt{1 - 2GM/r} dt$. Holding the clock static closer and closer to the horizon requires growing acceleration. The divergence of that acceleration concerns the static observer, whereas a freely falling observer has a regular local frame at the horizon.

The time coordinate is normalized so that it agrees with proper time for a stationary observer at infinity. Rescaling that coordinate rescales the corresponding Killing vector and therefore its surface gravity. This normalization issue returns in every comparison of black-hole temperatures.

Check. At the horizon the curvature invariant scales as $1/(G^4M^4)$. Increasing the mass makes the horizon less curved, even though the total gravitational mass is larger.

3 Crossing the horizon in regular coordinates

Define the tortoise coordinate by integrating

$$\frac{dr_*}{dr} = \frac{1}{1 - 2GM/r}, \quad r_* = r + 2GM \log \left| \frac{r}{2GM} - 1 \right|. \quad (3.1)$$

Radial null curves satisfy $t \pm r_* = \text{constant}$. Ingoing Eddington–Finkelstein time $v = t + r_*$ gives

$$ds^2 = - \left(1 - \frac{2GM}{r} \right) dv^2 + 2dvdr + r^2 d\Omega_2^2. \quad (3.2)$$

Every metric coefficient is finite at the future horizon. The coordinate singularity has disappeared without changing the curvature.

The outgoing radial null family obeys

$$\frac{dr}{dv} = \frac{1}{2} \left(1 - \frac{2GM}{r} \right). \quad (3.3)$$

Inside the horizon this derivative is negative. Even the outward-directed future null direction moves toward decreasing area radius. Timelike future directions lie inside the null cone, so they cannot escape either.

The maximal analytic extension contains additional exterior and white-hole regions. These are properties of the eternal solution. A black hole formed by stellar collapse does not inherit every region of that extension. Penrose diagrams should therefore be matched to the physical spacetime being discussed.

The logarithm in r_* is also the seed of the exponential redshift underlying Hawking radiation. It connects regular horizon coordinates to the late-time clock used by distant observers.

Check. The statement that a distant Schwarzschild clock assigns infinite coordinate time to horizon crossing does not imply infinite proper time along an infalling worldline.

4 Geodesics, photon orbits, and the ISCO

For an equatorial timelike geodesic, conserved energy and angular momentum per unit rest mass give

$$\dot{r}^2 + \left(1 - \frac{2GM}{r}\right) \left(1 + \frac{\ell^2}{r^2}\right) = E^2. \quad (4.1)$$

A circular orbit is an extremum of the effective potential. Differentiation gives

$$\ell^2 = \frac{GM r^2}{r - 3GM}. \quad (4.2)$$

The orbit is stable only when the second radial derivative of the potential is positive. This gives $r > 6GM$, with the innermost stable circular orbit at $6GM$.

For null geodesics the rest-mass term is absent, so the effective potential is proportional to $(1 - 2GM/r)/r^2$. Its extremum is at $r = 3GM$ and is unstable. The photon sphere is outside the event horizon; the two radii describe different physics.

At the timelike ISCO, substituting $r = 6GM$ into the circular-orbit energy gives $E = \sqrt{8/9}$. The difference from unity is the idealized binding energy

available before a particle leaves the sequence of stable circular orbits. It is not by itself a complete accretion efficiency model, which also depends on stress, radiation transport, and spin.

These calculations are useful checks on coordinate conventions. The area radius, not proper radial distance, appears in the quoted orbit locations. A plot against a different coordinate can shift visual features without changing the physical orbits.

Check. The three characteristic radii are $2GM$, $3GM$, and $6GM$: horizon, unstable circular photon orbit, and timelike ISCO.

5 Surface gravity and the near-horizon limit

For a static metric with $g_{tt} = -f(r)$ and $g_{rr} = 1/f(r)$, a simple horizon zero satisfies $f(r_h) = 0$ and $f'(r_h) \neq 0$. With the time normalization fixed,

$$\kappa = \frac{1}{2}f'(r_h) \quad (5.1)$$

for an outer horizon with positive derivative. For Schwarzschild, $f'(2GM) = 1/(2GM)$, so $\kappa = 1/(4GM)$.

Near the horizon, introduce proper radial distance ρ by

$$r - r_h = \frac{f'(r_h)}{4}\rho^2. \quad (5.2)$$

The two-dimensional time-radial metric becomes

$$ds^2 \simeq -\kappa^2\rho^2 dt^2 + d\rho^2. \quad (5.3)$$

This is Rindler space. The result explains why nonextremal horizons share a local thermal structure independently of many global details.

A useful invariant definition on a Killing horizon is $\xi^a \nabla_a \xi^b = \kappa \xi^b$. A constant rescaling of ξ rescales κ . In asymptotically flat space the normalization at infinity is natural; in AdS the chosen boundary metric fixes the normalization.

An extremal horizon has a double zero instead of a simple zero. The proper-distance expansion above does not apply, and its Euclidean geometry must be analyzed separately. Setting $\kappa = 0$ at the end of a nonextremal expression can miss a change in topology or limiting procedure.

Check. Surface gravity has dimensions of inverse length in the units used here. The physical temperature has the same dimensions once $\hbar$ and k_B are set to one.

6 Euclidean regularity and temperature

Continue $t = -i\tau$. The near-horizon metric becomes

$$ds_E^2 \simeq d\rho^2 + \kappa^2 \rho^2 d\tau^2. \quad (6.1)$$

Smoothness at $\rho = 0$ requires $\kappa\tau$ to be an angle of period 2π . The imaginary-time period is therefore

$$\beta = \frac{2\pi}{\kappa}, \quad T = \frac{\kappa}{2\pi}. \quad (6.2)$$

For Schwarzschild,

$$T = \frac{1}{8\pi GM}. \quad (6.3)$$

Restoring constants gives $T = \hbar c^3 / (8\pi G M k_B)$.

This argument identifies the temperature of the regular equilibrium Euclidean saddle. It does not alone specify the state of a black hole formed by collapse. The Hartle–Hawking state is thermal with both ingoing and outgoing flux in an eternal equilibrium setting, while the Unruh state describes an evaporating collapse geometry with outgoing Hawking flux and no corresponding thermal bath incident from past infinity.

A static local thermometer sees a Tolman-redshifted temperature,

$$T_{\text{local}}(r) \sqrt{-g_{tt}(r)} = T. \quad (6.4)$$

The divergent temperature assigned to static observers close to the horizon should not be confused with what a freely falling detector necessarily measures in a regular state.

Check. The periodicity is determined by eliminating a conical singularity. A factor-of-two error in the near-horizon radial coordinate produces a factor-of-two error in the temperature.

7 Hawking radiation as mode mixing

Positive frequency is defined relative to a time coordinate and a state. An outgoing mode that has fixed frequency for an observer far from a forming black hole has an exponentially changing frequency when traced backward toward the horizon.

The essential relation between a regular outgoing horizon coordinate and retarded time has the form

$$U \propto -e^{-\kappa u}. \quad (7.1)$$

Fourier decomposition across this relation mixes positive and negative frequencies. Analytic continuation around the logarithmic branch produces the thermal ratio

$$\frac{|\beta_\omega|^2}{|\alpha_\omega|^2} = e^{-2\pi\omega/\kappa}. \quad (7.2)$$

For bosonic modes, the Bogoliubov normalization then gives a mean occupation proportional to $(e^{\omega/T} - 1)^{-1}$, with $T = \kappa/(2\pi)$. Fermions obey the Fermi–Dirac distribution instead.

The spectrum observed at infinity also contains transmission probabilities through the exterior curvature potential. These greybody factors prevent the flux from being an exact blackbody spectrum at every frequency.

The derivation assumes quantum fields on a semiclassical background and a suitable regular short-distance state. The familiar trans-Planckian tracing of late-time modes motivates scrutiny of ultraviolet assumptions; it is not, by itself, a calculation of quantum-gravity corrections or a proof that the thermal result fails.

Reading connection: Hawking’s original derivation is [3]. Harlow [5] develops the quantum-information interpretation and the assumptions behind the information problem.

8 The first law and area entropy

For stationary black holes in Einstein gravity, variations among nearby solutions obey

$$dM = \frac{\kappa}{8\pi G} dA + \Omega_H dJ + \Phi_H dQ. \quad (8.1)$$

Comparing the coefficient of dA with $T = \kappa/(2\pi)$ identifies

$$S = \frac{A}{4G}. \quad (8.2)$$

With constants restored, the entropy is $k_B A c^3 / (4G\hbar)$. The formula is dimensionless in the units used throughout the calculations.

For Schwarzschild, $A = 16\pi G^2 M^2$, so

$$S = 4\pi G M^2, \quad T \frac{dS}{dM} = 1. \quad (8.3)$$

This is a direct consistency check of the numerical coefficient. The first law relates variations between equilibrium solutions; it should not be interpreted as an arbitrary local evolution equation during violent collapse.

Classical horizon area increases under suitable energy and global assumptions. Hawking radiation violates the simple classical matter assumptions needed for that theorem, so an evaporating black hole can shrink. The generalized second law instead involves horizon entropy together with the entropy outside.

In higher-derivative gravity, black-hole entropy is generally a Noether-charge expression rather than area alone. Inserting $A/(4G)$ into a modified action without checking its entropy functional can give an incorrect first law.

Reading connection: Bekenstein's entropy proposal is [4]; Wald's generalization is [6].

9 Negative heat capacity and evaporation

Differentiating the Schwarzschild temperature gives

$$C = \frac{dM}{dT} = -8\pi G M^2. \quad (9.1)$$

An isolated black hole becomes hotter as it loses mass. A Schwarzschild black hole in an infinite heat bath at the same temperature is canonically unstable: a small upward mass fluctuation makes it colder and encourages further absorption.

A rough four-dimensional radiation estimate gives power proportional to area times T^4 . Since area scales as $G^2 M^2$ and temperature as $1/(GM)$,

$$-\frac{dM}{dt} \propto \frac{1}{G^2 M^2}, \quad t_{\text{evap}} \propto G^2 M^3. \quad (9.2)$$

The coefficient depends on particle species, greybody factors, and mass thresholds. It is not fixed by dimensional analysis.

Integrating a mass-loss law proportional to $-M^{-2}$ gives a mass cubed that decreases linearly with time. This approximation is useful while the black hole is large and evolves slowly compared with its light-crossing time. It fails near the endpoint, where curvature and backreaction can invalidate semiclassical reasoning.

A reflecting cavity or anti-de Sitter boundary can alter the stability question. Stability belongs to a specified ensemble and boundary condition, not to an isolated formula for temperature. The same family of solutions can have a useful microcanonical interpretation even when its canonical heat capacity is negative.

Check. “Black holes have negative heat capacity” is too broad. Large AdS black holes provide an elementary counterexample.

10 Charge, rotation, and extremality

In geometrized units for charge and mass, the Reissner–Nordstrom metric has

$$f(r) = 1 - \frac{2M}{r} + \frac{Q^2}{r^2}, \quad r_{\pm} = M \pm \sqrt{M^2 - Q^2}. \quad (10.1)$$

The outer-horizon temperature is

$$T = \frac{r_+ - r_-}{4\pi r_+^2}. \quad (10.2)$$

It vanishes in the extremal limit $|Q| = M$. These formulas use geometrized M, Q in this section; comparison with the earlier Schwarzschild expressions requires restoring the relevant factors of G and the electromagnetic normalization.

For Kerr, with $G = 1$ and $a = J/M$,

$$r_{\pm} = M \pm \sqrt{M^2 - a^2}, \quad \Omega_H = \frac{a}{r_+^2 + a^2}, \quad (10.3)$$

$$T = \frac{r_+ - r_-}{4\pi(r_+^2 + a^2)}. \quad (10.4)$$

The horizon generator is a combination of time translation and rotation. A bosonic wave can extract rotational energy when its frequency satisfies the superradiant condition $0 < \omega < m\Omega_H$, with the usual azimuthal mode number m . Charged fields add the electrostatic contribution.

Superradiant amplification does not automatically imply an instability. Repeated amplification requires a trapping or reflecting mechanism. Likewise, an extremal entropy is not automatically a thermal entropy at a nonzero temperature: ground-state degeneracy and limiting ensembles must be distinguished.

Check. The inner horizon is not a second ordinary event horizon for an exterior observer. Its stability and predictability properties are substantially different.

11 AdS black holes and ensemble stability

For a four-dimensional Schwarzschild–AdS solution,

$$f(r) = 1 + \frac{r^2}{L^2} - \frac{2GM}{r}. \quad (11.1)$$

The horizon condition yields

$$M = \frac{r_h}{2G} \left(1 + \frac{r_h^2}{L^2} \right), \quad T = \frac{1 + 3r_h^2/L^2}{4\pi r_h}, \quad S = \frac{\pi r_h^2}{G}. \quad (11.2)$$

The temperature has a minimum at $r_h = L/\sqrt{3}$. Small black holes below this radius have negative heat capacity; larger ones have positive heat capacity.

Relative to thermal AdS with the same boundary temperature, the free energy is

$$F = M - TS = \frac{r_h}{4G} \left(1 - \frac{r_h^2}{L^2} \right). \quad (11.3)$$

It changes sign at $r_h = L$, corresponding to $T = 1/(\pi L)$. This is the Hawking–Page transition in this four-dimensional bulk example.

There is a range of locally stable black holes with positive free energy, so local stability and global thermodynamic dominance are distinct. One must

compare saddle actions at fixed boundary data, not merely ask whether a black hole exists.

The boundary spatial sphere has a finite radius proportional to L in the chosen conformal frame. The associated scale allows a finite-temperature transition. A planar conformal boundary has different thermodynamics, as discussed in the AdS/CFT note.

Reading connection: The original phase-transition analysis is Hawking and Page [7].

12 BTZ and a microscopic entropy check

Three-dimensional Einstein gravity with negative cosmological constant admits the BTZ black hole. For the rotating solution,

$$M = \frac{r_+^2 + r_-^2}{8G_3L^2}, \quad J = \frac{r_+r_-}{4G_3L}, \quad S = \frac{2\pi r_+}{4G_3}. \quad (12.1)$$

There are no local propagating gravitons in pure three-dimensional Einstein gravity, but global identifications and boundary degrees of freedom remain physically nontrivial.

The Brown–Henneaux central charge is $c = 3L/(2G_3)$. Write the left and right cylinder excitation energies as $L_0 - c/24 = (ML + J)/2$ and $\bar{L}_0 - c/24 = (ML - J)/2$. In a regime where the Cardy approximation is valid,

$$S = 2\pi\sqrt{\frac{c}{6}\left(L_0 - \frac{c}{24}\right)} \quad (12.2)$$

$$+ 2\pi\sqrt{\frac{c}{6}\left(\bar{L}_0 - \frac{c}{24}\right)}. \quad (12.3)$$

Substitution reproduces $2\pi r_+/(4G_3)$. The vacuum-energy shifts are essential for using the cylinder spectrum consistently.

The agreement connects a geometric horizon area with a density of states in a two-dimensional conformal theory. It is not a state-by-state construction of a generic interior. The range of Cardy universality must also be checked; a large central charge by itself does not justify every low-energy extrapolation.

Check. Set $J = 0$, so $r_- = 0$. The left and right contributions are equal, and each supplies half the entropy.

Reading connection: The BTZ solution is [8]; the asymptotic symmetry analysis is [9].

13 String microstates: what is counted

A landmark string-theory calculation counts protected states carrying D-brane charges and momentum in a supersymmetric setting. In the familiar D1–D5–momentum description, the leading entropy behaves as

$$S = 2\pi\sqrt{Q_1 Q_5 n} \quad (13.1)$$

in the corresponding large-charge regime and conventional charge normalization.

At weak coupling, the counting can be organized through an effective two-dimensional theory. At stronger coupling, the same charges support a black-hole geometry with macroscopic area. Supersymmetry protects suitable quantities as the coupling changes, allowing a comparison across regimes where neither description is simultaneously elementary.

The conceptual achievement is a microscopic explanation of the leading area entropy for specific black holes. It does not establish that the same protected counting procedure is directly available for an uncharged four-dimensional Schwarzschild black hole. Nor does a degeneracy formula by itself tell us what an individual microstate looks like to an infalling observer.

Care is required about indices versus absolute degeneracies. A supersymmetric index includes signs and can exhibit cancellations. In successful applications, one must explain why its leading growth captures the black-hole entropy in the regime being considered.

Subleading corrections test more than a leading square-root formula. They can involve higher-derivative terms, quantum fluctuations, and ensemble transformations. Such comparisons are meaningful only when the charges, ensemble, and normalization are fixed consistently.

Reading connection: Strominger and Vafa’s original microscopic entropy result is [10].

14 The information problem as an entropy conflict

Suppose a black hole forms from a pure state and evaporates completely into radiation, with unitary evolution and no inaccessible remnant sector. Then the final radiation state must be pure. Its fine-grained von Neumann entropy must eventually return to zero.

The leading semiclassical Hawking calculation instead produces outgoing radiation entangled with interior partners. Repeating that approximation gives an entropy that grows as more radiation is emitted. The conflict concerns the fine-grained state, not whether the average energy spectrum looks thermal.

A pure state can have locally thermal reduced density matrices. Therefore recovering unitarity need not require large visible deviations from a thermal spectrum in every early Hawking quantum. Information may reside in correlations that are invisible to simple one-body observables.

The Page model estimates the entropy of a smaller subsystem in a typical pure state of a larger bipartite Hilbert space. At leading order it follows the logarithm of the smaller effective dimension. Applied heuristically to evaporation, this gives a rising and then falling curve. It is a kinematic typicality model, not a derivation of the evaporation dynamics.

A precise discussion should specify whether entropy is coarse grained, fine grained, ensemble averaged, or evaluated in a fixed microscopic state. These quantities can differ even when they share an early-time approximation.

Check. A thermal density matrix and a pure state with thermal small subsystems have different global entropies. The similarity of a few correlators cannot decide between them.

Reading connection: For the structure of the paradox and quantum-information preliminaries, see [5].

15 Generalized entropy and quantum extremal surfaces

The generalized entropy combines a geometric contribution with quantum entropy across a candidate surface:

$$S_{\text{gen}} = \frac{A}{4G} + S_{\text{bulk}}. \quad (15.1)$$

Both terms require a compatible renormalization prescription. Ultraviolet divergences in bulk entropy are absorbed together with gravitational coupling renormalization and, when needed, higher-curvature counterterms.

A quantum extremal surface extremizes generalized entropy. Among admissible competing extrema, one selects the entropy prescription appropriate to the state and boundary region. Extremization comes before comparison of saddle values; an arbitrary surface with a small area is not a valid candidate.

For radiation in suitable gravitational models, the island prescription takes the schematic form

$$S(R) = \min_I \text{ext} \left[\frac{\text{Area}(\partial I)}{4G} + S_{\text{matter}}(R \cup I) \right]. \quad (15.2)$$

The no-island and island saddles can exchange dominance. This produces a Page-like turnover in controlled examples while leaving many local semiclassical predictions intact.

An island does not represent an independently duplicated copy of radiation information. It signals that the entropy calculation and reconstruction region are organized differently once gravitational constraints and competing saddles are included.

Replica-wormhole derivations introduce further questions about ensembles and factorization in some models. The existence of successful Page-curve calculations should not be promoted into a complete microscopic description of all realistic evaporation processes.

Reading connection: Quantum extremal surfaces are introduced in [11]; a detailed review of radiation entropy and islands is [12].

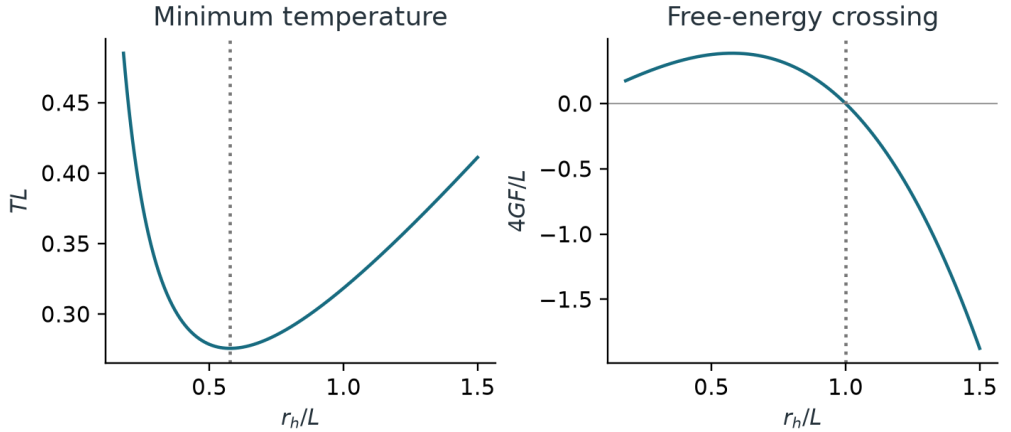


Figure 1: Four-dimensional Schwarzschild–AdS thermodynamics from the explicit formulas in Section 11. The temperature minimum and free-energy crossing occur at different radii. The latter determines global dominance relative to thermal AdS.

16 Worked example: checking the first law in AdS

Use the Schwarzschild–AdS formulas in Section 11. Differentiation gives

$$\frac{dM}{dr_h} = \frac{1}{2G} \left(1 + \frac{3r_h^2}{L^2} \right), \quad \frac{dS}{dr_h} = \frac{2\pi r_h}{G}. \quad (16.1)$$

Multiplying the entropy derivative by the temperature reproduces the mass derivative exactly. This verifies $dM = TdS$ at fixed L .

The heat capacity follows by dividing by

$$\frac{dT}{dr_h} = \frac{1}{4\pi} \left(\frac{3}{L^2} - \frac{1}{r_h^2} \right). \quad (16.2)$$

Thus

$$C = \frac{2\pi r_h^2}{G} \frac{1 + 3r_h^2/L^2}{3r_h^2/L^2 - 1}. \quad (16.3)$$

The divergence marks the minimum temperature, where two branches meet. It is not the Hawking–Page point, which occurs at $r_h = L$ and is found by comparing free energies.

If the cosmological constant is varied, the first law acquires an additional term in the extended-thermodynamics interpretation. That is a different thermodynamic question from varying the mass of a black hole in a theory with fixed AdS radius. Keeping L fixed here makes the comparison with a fixed boundary field theory straightforward.

This worked example separates three operations often mixed together: differentiating a solution family, diagnosing local stability, and comparing global saddle dominance. Each uses the same equations but asks a different question.

17 Worked example: redshift and detector temperatures

For a Schwarzschild black hole, a static observer at radius r has local temperature

$$T_{\text{local}} = \frac{1}{8\pi GM \sqrt{1 - 2GM/r}}. \quad (17.1)$$

At $r = 4GM$, the redshift factor is $1/\sqrt{2}$, so the local equilibrium temperature is $\sqrt{2}$ times the temperature assigned at infinity.

The proper acceleration of a static observer is

$$a = \frac{GM}{r^2 \sqrt{1 - 2GM/r}}. \quad (17.2)$$

Near the horizon, $a/(2\pi)$ approaches the same leading divergence as T_{local} . Away from the horizon the two formulas are not identical. The near-horizon Rindler approximation explains the local agreement, while the global geometry and state determine the complete detector response.

A detector on a geodesic is a different physical experiment. Its response depends on the field state, trajectory, switching function, and energy gap. Replacing its trajectory with a static one while keeping the same words “near the horizon” changes the question.

The lesson is to specify both observer and state before assigning a measured temperature. Euclidean periodicity fixes an equilibrium correlation function; the Tolman relation concerns static observers in that equilibrium; an evaporating state and a freely falling detector require additional analysis.

Check. At very large r , the static acceleration tends to zero while the Hartle–Hawking bath temperature tends to the finite asymptotic Hawking temperature. This immediately shows that the two expressions cannot agree everywhere.

18 Exercises with solution outlines

- 1. Curvature at the horizon.** Substitute $r = 2GM$ into the Kretschmann scalar. The result is $3/(4G^4M^4)$. This verifies both finiteness and the inverse-fourth-power mass dependence. The curvature becomes Planckian only when the mass approaches the corresponding microscopic scale.
- 2. Photon sphere.** Differentiate $(1 - 2GM/r)/r^2$. The derivative vanishes at $r = 3GM$; the second derivative there is negative. The circular null orbit is therefore unstable. This instability is a property of a geodesic orbit and should not be equated automatically with the many-body quantum Lyapunov exponent of a dual theory.
- 3. Entropy and temperature.** Starting only from $S = 4\pi GM^2$ and the first law with no charge or rotation, invert dS/dM to recover $T = 1/(8\pi GM)$. Differentiating again gives negative heat capacity.
- 4. Evaporation scaling.** Integrate a mass-loss rate proportional to $-M^{-2}$. The evaporation time scales with the initial mass cubed. Explain why the proportionality coefficient is not universal and why the last stage lies outside the approximation.
- 5. Competing AdS saddles.** At $r_h = L/\sqrt{3}$, the heat capacity diverges but $F > 0$. At $r_h = L$, $F = 0$ and the heat capacity is positive. This demonstrates a locally stable but globally subdominant branch between those radii.
- 6. Purity.** Construct a Bell pair and trace out one qubit. The reduced entropy is $\log 2$, while the full entropy is zero. The example is the simplest algebraic model of why radiation can look mixed when its purifying degrees of freedom are omitted.

19 What remains to be understood

Classical causal structure, the temperature of a nonextremal horizon, and the leading semiclassical radiation calculation are well-defined within their stated frameworks. Their synthesis into a complete account of evaporation requires more than extrapolating any one framework beyond its domain.

Microscopic entropy counts in supersymmetric string constructions and entropy-saddle calculations in gravitational models provide different kinds of evidence. The former constrain densities of protected states; the latter constrain fine-grained entropy in semiclassical path integrals. Neither should be described as a direct laboratory observation of an individual black-hole interior.

For further study, track a single observable across descriptions. The thermal partition function leads from Euclidean saddles to density of states. A two-point function leads from wave propagation to quasinormal modes. Radiation entropy leads from reduced density matrices to replica geometries. An out-of-time-order correlator leads from scattering near a horizon to scrambling.

Keeping those observables distinct makes the relation to high-energy theory precise. The entropy of a region, a geodesic instability, a thermal relaxation time, and a scrambling time are not interchangeable, even when the same black-hole geometry appears in each calculation. The companion notes develop the corresponding boundary observables and their limitations.

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